

Whether we are discussing individual plasmonic nanostructures, or metamaterials and photonic crystals composed of dielectric or metallic building blocks, structural designs play a central role. Therefore, we have to rely on the second approach: electromagnetic modelling based on numerical simulation methods as the finite-difference time-domain method, the finite-element method, the finite integration technique or the method of moments, with or without optimization algorithms. This procedure largely relies on past experience of the design templates, and owing to constraints on simulation power and time, only limited design parameters are adjusted in searching for the optimal structure. The inverse design problem, meaning the direct retrieval of the proper structure for the desired optical performance, exploration of a much larger degree of freedom in the design space, and hence is even more challenging. To search the formidably large design space efficiently, the inverse design procedure is usually guided by optimization either gradient-based approaches (for example, topology optimization, adjoint method or level-set method) or approaches (such as genetic algorithms or particle swarm algorithms). Such inverse design algorithms enable one to find non-intuitive, irregularly shaped photonic structures that outperform designed structures in many applications, such as silicon photonic components, photonic crystals and metamaterials. Readers interested in inverse design in photonics can refer to recent reviews on this topic^{9,11}. Deep learning allows a computational model composed of multiple layers of processing units to learn multiple levels of information in given data¹². In light of its exceptional success in domains related to computer science and engineering, including computer vision¹³, language processing¹⁴, speech recognition¹⁵, knowledge graphs¹⁶ and decision making¹⁷, deep learning has attracted attention from researchers in other disciplines, including materials science¹⁸, chemistry¹⁹, laser physics²⁰, particle physics²¹, mechanics²², computational imaging²³ and microscopy²⁴, demonstrating potential to circumvent the drawbacks of traditional methods and create unprecedented opportunities in these areas. The unique advantages of deep learning lie in its data-driven methodology, which allows the model to discover useful information automatically from a huge amount of data, in sharp contrast to physics- or rule-based approaches. Over the past few years, deep learning has become a radically new approach in the context of photonic design. In this Review, we draw attention to a collection of recent results that showcase the power of deep learning in the design of structures, materials and devices, where empirical or traditional approaches are infeasible or inefficient. We begin by providing background on the deep-learning model for photonics and highlighting the formulation, and advantages of deep neural networks. Then we discuss several major model architectures, from the basic multilayer perceptron (MLP) and advanced deep neural to hybrid models with other optimization methods, emphasizing their potential to design photonic crystals, metamaterials, plasmonic nanostructures and integrated silicon photonic devices (Fig. 1).

Background

Historically, deep learning can be traced back to the 1940s, and it went through many different names before becoming known by this term²⁵. Therefore, deep learning was known by the term artificial neural networks (ANNs) since the 1980s, accompanied by the wave of AI research, which largely emerged through a movement called connectionism²⁶. The modern era of neural network research began with a breakthrough in 2006: Hinton, who later coined the term deep learning, showed that deep neural networks could be efficiently trained using a strategy called greedy layer-wise pretraining²⁸. Deep learning then gained broad popularity, as researchers were able to train deeper neural networks than had been before, and the importance of depth of the model architecture was theoretically realized. Aided by the ever-increasing scale of available data, deep learning still leads the current boom in AI research, with the surpassing previous models by a large margin, or even beating humans²⁹. In the context of photonics research, the introduction of deep learning (or more appropriately ANNs, at that time) dated to the 1990s during the second prevalence period of AI research. The use of ANNs in microwave and other photonic designs follows a standard supervised learning paradigm, which aims to find a mapping between input variable X and output variable Y . The training process is to optimize the model parameters to make it an accurate representation of the desired mapping, a previously collected training dataset containing examples of X and its corresponding label Y . Starting with random initialization, this data-driven learning approach modifies the model parameters iteratively on the dataset according to a specific loss function evaluated on true labels Y , until a convergence occurs and the generalized model is able to predict unseen data. With well-understood physics, mature simulation tools and thus little difficulty in data acquisition, finding the relations between these variables is less complicated for an MLP model. On the other hand, more complex photonic designs such as plasmonic nanostructures, metamaterials, photonic crystals or photonic devices were still in their infancy, with limited understanding in the community. These structures and devices, featuring deep sub-wavelength dimensions, large design flexibility, extreme dispersions or complex performance characteristics, were not tractable to the shallow ANN model at that time. Related research then stagnated until the recent epoch-making development of deep learning. The past decade has witnessed the rise of deep learning with unprecedented impact on a plethora of research topics. With the invention of new training and regularization techniques such as ReLU (rectified linear units) activation³⁰, and batch normalization³⁸, it is now feasible to design and train deeper and deeper neural networks that can exploit larger information with better performance. The photonics community also benefited from the rapid advances in deep-learning techniques. Unlike rule-guided optimization that explores the design space by following certain strategies case by case, deep learning, a data-driven method, aims to describe the design space holistically, using the training data as samples. Therefore, with generalization ability within a given design space, deep learning can produce fast and accurate designs without the need for case-by-case, time-consuming numerical calculations. Well-trained deep-learning models can directly set up a mapping from design to optical properties of target photonic structures and vice versa. In addition, deep learning can interact with traditional optimization methods to improve the algorithm performance. In this section, we will discuss some relevant model architectures and their applications in solving photonics problems. In modern deep-learning models with sophisticated and task-specific architectures, MLP often serves as a bottleneck layer to extract meaningful features as a compact representation of high-dimensional data such as a fully connected neural network. An L -layer consists of one input layer, one output layer and $L-2$ hidden layers. Even before the astounding success of deep learning in computer vision³³, researchers explored MLP in photonics-related problems following the design schemes devised by the microwave community. The potency of these models was largely limited by immature training strategy, lack of data and thus shallow model. In 2018, Itzik Malkiel and co-workers reported a bidirectional MLP-based deep-learning model that could be used to design plasmonic nanostructures⁴³. In an advance on earlier work, the researchers managed to model the intricate physical relationship between photonic structures and their optical characteristics by resorting to a deeper network configuration. The scale of the data-set, input complexity and the depth of the model show that, from a data-driven view, deeper models with larger datasets are a potential alternative approach to solve photonic design tasks. Meanwhile, to deal with specific photonic problems, adaptation of the MLP-based model burgeoned, with improvements in the model architecture, training strategy and application. Wei Ma and co-workers reported a deep-learning model with two bidirectional neural networks that were integrated by an ensemble learning strategy to achieve on-demand design of chiral metamaterials, which exhibit distinct responses when light is left-circularly polarized (LCP) or right-circularly polarized (RCP)⁴⁴. When trained on large datasets, MLP-based models for inverse design often fail to converge, since there exist multiple information satisfying similar requirements but with very different designs. By indirectly training in this tandem configuration, the inverse retrieval outputs are forced to converge to only one solution guided by the forward model, efficiently solving the data inconsistency issue that arises from the fundamental property of uniqueness in the inverse design problem. John Peurifoy and colleagues used a single neural network to approximate light scattering by multi-layer nanoparticles, the analytical gradient obtained from the model was used for structural optimization given specific requirements in the deep neural network and interior-point methods, a numerical inverse design algorithm suitable for nanoparticles. Following these first forays into applying data-driven methodology in photonics research, deep MLP models and their applications have been extensively studied for many photonic design tasks, including topological photonics⁴⁷, integrated silicon

colour generation from nanostructures^{49,50}, metamaterials^{51,53}, plasmonic structures⁵⁴ and photonic crystals⁵⁵. 2d illustrates topological photonic structures whose edge-state dispersion is well reconstructed by an MLP model⁴⁷. 2e, the researchers propose to use deep learning to design silicon-on-insulator-based 1×2 power splitters with various transmission ratios at two ports (T^1 and T^2), in which the device structure is represented by a matrix of binary variables, indicating whether the silicon is etched or not at certain coordinates⁴⁸. In addition to design problems, researchers have also developed sophisticated MLP models to tackle other problems in such as modal classification and effective refractive index retrieval in waveguides^{56,57}, dimension reduction to reveal physics⁵⁸ and near-field manipulation of plasmonic nanoantennas⁵⁹. As the plain form of the deep-learning model, MLP requires vectorized input and output. By parameterizing model inputs and outputs into vectors composed of several discrete and dimensionless elements, the model can be adapted to a broad range of photonic applications regardless of the underlying physics. Advanced deep-learning techniques. Although MLP models offer an effective approach for many photonic tasks, the simple connections in the model still pose some difficulties when the structure, target response and design space are either multimodal or hard to parameterize. Therefore, application-specific model architectures have been invented for photonic research or transplanted from other including CNNs specialized to input with local semantic correlation, RNNs for time-dependent inputs and deep generative to deal with structured-output conditions. Such advanced models can extend the functionality and improve performance in many photonic design problems, which discussed below.

Convolutional neural networks. In deep learning, a CNN is a TCNNs are commonly implemented as a part of generative models for the discovery and design of photonic structures large degrees of freedom. The basic architectures of CNN- and TCNN-based generative models that transform data between high-dimensional and low-dimensional feature vectors are illustrated in the lower panels of Fig. The convolution operation in a CNN is translationally symmetric, making it suitable to model periodic photonic such as photonic crystals or metamaterials. These network structures can be used to train surrogate models for the simulation of complex photonic structures with degrees of freedom⁶⁰.

For photonic design, RNNs are suitable to model optical signals or spectra in the time domain with a specific line shape from various modes of the resonance. Network systems hybridizing CNNs and RNNs are promising techniques to model and design photonic devices that unconventional spatiotemporal properties of light. Unlike their deterministic counterparts, such models can handle one-to-many mapping and produce structured outputs prescribed requirements, as illustrated in Fig. Empowered by deep-learning algorithms, deep generative models can produce data that replicate in the same way as or similar to the training dataset. AEs are used to reduce the dimensionality of the design space and optical response features of photonic devices, providing information on the light-matter interaction for device optimization⁶⁰. Because the latent space is a continuous representation of the training data, new designs can be constructed by sampling latent space⁷⁰.

Leveraging the reduced dimension and continuity of the latent space, Wei Ma and colleagues used a VAE-based model to encode the meta-atoms of double-layered chiral metamaterials and their optical responses, enabling the of the complex relationship between structure and performance without extensive data collection⁷¹. More importantly, the model can solve the one-to-many mapping inverse problem and come up with distinct structures to satisfy the same requirement of chiroptical responses. 3c (bottom panel), is another class of deep generative model constructed with a generator and a discriminator⁴⁰. With the ability to generate massive nanostructures within short time, GANs have been used in the design and of dielectric and metallic metasurfaces in a stochastic manner⁷². Recently, Zhaocheng Liu and co-workers have proposed a framework leveraging GANs to inversely design metasurface that match on-demand design objectives⁷³.

3e, given a user-defined transmittance spectrum for the inverse design problem (left panel), the framework can identify a with a spectral behaviour that matches the input one with high fidelity (right panel). Besides the standard architecture of the GAN and VAE, some other varieties of generative models can also be for the discovery and design of photonic structures. For instance, compositional pattern-producing networks (CPPNs) have been reported as serving as a generator in the GAN to produce high-quality nanostructure patterns for the inverse design of metasurfaces with multiple meta-atoms in a unit. Other examples of generative models applied in photonics research have also been reported. For example, U-net, a deep neural network architecture originally designed for image segmentation, has been used to predict the near-field optical responses of arbitrary 3D nanostructures, enabling the approximation of various far-field of nanostructures⁷⁵.

By combining traditional optimization techniques and a generative model, global optimization of a 1D meta-grating can achieved with high efficiency for various wavelengths and deflection angles⁷⁶. Optimization assisted by deep-learning algorithms. Recent By hybridization of conventional optimization methods with advanced deep-learning algorithms, it is possible to exploit best aspects of both approaches and greatly reduce the required computational resources. 4b, while GAN-generated designs show similar performance to those obtained from the direct topology optimization, assisted optimization generates five times as many high-performance devices in the same time. More recently, an adversarial autoencoder (AAE), consisting of three neural networks (encoder, decoder and discriminator) has been coupled with a topology optimization framework for the development of a thermal emitter design⁸². Along with efficient optimization search, such an approach promises unparalleled control over the data distribution the compressed design space. Recently, linear principal component analysis has been used to map and characterize a multiparameter design space of a system⁸³.

4d, the developed approach consists of three main steps: (1) construction of a sparse collection of pre-optimized designs applying conventional optimization methods; (2) determination of a lower-dimensional sub-space of highly efficient by applying a dimensionality reduction algorithm to the high-dimensional parametric space of pre-optimized designs; and (3) exhaustive mapping of the design sub-space to determine the continuous region of highly efficient designs. Such global analysis ensures efficient multiparametric optimization and reveals the limitations of performance or structure the given design configuration. The feasibility of using data-driven methods for solving partial differential equations has been investigated very recently, the demonstration of learning an optimal approximation of derivatives needed for time-domain simulations⁸⁵ or using networks for solving partial differential equations^{86,87}.

Discussions and perspectives In this section, we will comment on the directions, challenges and perspectives of interfacing deep learning with photonics. Deep learning for complete control of light. It is believed that deep learning can serve as a powerful tool in finding complex relations between a structure and its optical responses. Spectra can be conveniently discretized into vectors⁴³ (single spectrum) or matrices⁴⁴ (multiple spectra), and thus can be incorporated into the deep-learning model. To aid in the complete control of light, one immediate task that we need to work on is to enhance the capability of deep-models with more degrees of freedom, such as the phase⁴, angular momentum, trajectory, nonlinearity, topology and field distributions⁷⁵.

Reinforcement learning is another approach worthy of further exploration, to expand the photonic design capability⁸⁰. This indirect optimization process balances the current knowledge and unknown domains, and could potentially invest more optical properties of photonic design tasks to realize complete control of light. In these circumstances, unsupervised and semi-supervised learning strategies can be leveraged to alleviate the burden of collection. Unlike supervised learning, unsupervised learning algorithms, such as principal component analysis, and AEs/VAEs/GANs, only a small number of labels, usually serving as tools for data clustering and dimensionality reduction. The processed data from unsupervised learning strategies can be analysed to unveil the structural parameters of the devices that have most impact, assisting the neural network to recognize the essential information of the design without

data collection^{90,91}.

Massive data collection can also be mitigated by merging the deep-learning models with underlying physics.

For example, instead of training a model with enormous datasets to approximate the bidirectional relationship between structures and their responses, deep learning can be adopted as intermediate steps to efficiently solve partial differential equations that describe the physical systems^{87,88,92}.

Alternatively, learning the governing laws behind the physical phenomena can also reduce the dependence on data^{86,93}.

Deep learning in this strategy is usually used to extract features from limited data for the regression of a few parameters that parametrize the models or equations of the system.

Furthermore, it is possible to generate a solution that satisfies certain partial differential equations by physics-informed learning models⁹⁴.

In other circumstances where a neural network is well trained for the approximation of certain physical processes, transfer learning can be applied to migrate the knowledge of the neural network to other similar simulation scenarios with substantially less data collection⁹⁵.

Comparative advantages of deep learning over conventional optimization

From the viewpoint of efficiency, a well-trained deep-learning model is faster by orders of magnitude than traditional algorithms⁴⁶, on the premise that enough training data are collected in advance.

The difference between data collection for deep learning and that for numerical optimization algorithms lies in how they make use of data.

For deep learning, the model actively accumulates training data that holistically describe the problem under investigation, that the model is driven by data in the sense that the quantity and quality of pre-collected training data contribute to its performance.

Usually, for a specific task, the quantity of data required by deep learning is much more than produced during a single optimization.

However, data collection for deep learning is a one-time cost, as opposed to an ongoing cost as in the case of *ab initio* optimization.

This feature makes deep learning especially competitive in photonic design tasks of periodic structures such as photonic crystals, where a huge number of unit cells need to be optimized to constitute a device or a system.

Meanwhile, balancing training time and run time for a deep-learning model requires efficient exploitation of data, probably in an online manner where data are continuously provided as the model evolves according to different tasks.

To some extent, deep learning has similarities with gradient-based optimization methods, because the optimization of the learning model is also directed by the average gradient of the loss function with respect to all the training data.

On the other hand, similar to gradient-free optimization that introduces more stochasticity to enlarge the possible design space, deep learning uses a large amount of pre-collected training data to extend the scope of design varieties far beyond what is possible with optimization methods.

In conjunction with traditional numerical optimizations, a fast deep-learning approximator for the following optimization can be constructed by saving the simulation data right from the beginning of the optimization^{95,96}.

On the other hand, deep learning can help to identify a global minimum in various ways.

For example, fast inference speed enables global search algorithms for the design of highly complex structures.

In more practical cases, traditional optimization and data-driven methods can be jointly incorporated into the same design line, where deep learning serves to identify solutions near global minimum and traditional optimization refines them to the extreme.

Even if a true global optimum can hardly be found in a complex photonic design task, these optimization methods, including deep learning, always yield fairly good results beyond empirical designs.

In some cases in which only a small amount of data can be collected in a reasonable time, incorporating statistical learning or other machine-learning algorithms to the design strategy is also advantageous.

However, because deep-learning and machine-learning models always require sufficient training datasets, data-driven methods may not be an effective starting point in extreme instances where retrieving a single data point takes from hours to days.

Instead, direct optimization such as Bayesian optimization could be considered before deep-learning models are applied.

Enlarging design space for global photonic designs. Addressing

Thus far, optical technologies have mainly used photonic optimization in a restricted design space that is largely limited to topology (geometry) and shape.

Such optimization often omits in-depth feedback from other critical design layers including the evolution of electromagnetic properties with time or in harsh environments, as well as fabrication and characterization constraints.

Global optimization search within the extended compressed space will not only ensure optimization of the design topology but will also provide optimal material properties and composition, or even guide the fabrication process to achieve the best performance of the targeted photonic device, architecture and system.

We envisage that the global optimization framework could become an essential part of a more generic, hierarchical learning framework based on a multistep strategy.

By using the available databases on chosen material platform properties, machine-learning-driven global optimization can then be used as the third step to obtain ultimate-efficiency device designs and to determine suitable fabrication (growth, doping level, stoichiometry and so on) and integration schemes.

While the rapid advances in deep learning provide radically new approaches to solve photonic design problems, this is not one-way but interactive⁹⁹.

Big data, the catalyst of deep learning, continues to revolutionize AI research with record-breaking improvements in various domains, while at the same time it causes an exponential increase in computational power and energy consumption.

At present, most deep-learning algorithms are deployed in conventional computers with von Neumann architecture, whose serial nature is an intrinsic barrier to efficient support for neural networks.

Even with some application-specific integrated circuits that are deliberately optimized to run deep-learning models, most current models for practical applications still take a formidable time and cost to train.

Superior to their electronic counterpart in both speed and power consumption, photonic platforms for deep learning, including nanophotonic scatterers¹⁰⁰, integrated silicon photonic chips^{101,103} and 3D-printed diffractive layers¹⁰⁴, are under active development.

The tremendous acceleration of deep-learning models on an optical platform depends on the capability of parallel signal processing of light, which makes matrix-vector multiplications in constant time with respect to the matrix dimension, in contrast to the quadratic time complexity on a digital processor.

The interactions between new photonic structures and deep learning may overcome the limitation of current computing architectures and systems, and potentially lead AI research to new horizons.

Indeed, it has been demonstrated that inverse-designed metastructures can solve integral equations using electromagnetic simulations. The potential of new photonic structures, some of which may be designed by deep learning, to enable unconventional and AI techniques is worthy of further exploration.

summary

The concept of deep learning has gone far beyond being a computational analogy of biological neural systems and has become a powerful tool that solves extremely complex problems by building up multilevel abstraction of massive data.

For the photonics community, deep learning and other AI techniques are currently transforming the areas of optical design, device integration and measurements.

Deep-learning techniques have already demonstrated their tremendous potential for photonic structure design, device architecture, materials and entire optical systems, and will continue to uncover new ways to unparalleled speed-up of measurements and even unlocking new optical effects.

In this Review, we have surveyed various model structures for photonic design, ranging from individual plasmonic structures to metamaterials comprising an array of meta-atoms, and to integrated photonic devices.

Deep learning and AI researchers should team up with optical scientists to develop unorthodox, physics-driven algorithms and networks that are not only robust, generative and interpretable while using fewer data, but also provide unconventional ways to realize unparalleled optical functionalities.

Such cross-disciplinary approaches merging AI, photonics and materials platforms will allow for large-scale photonic devices with unique functionalities as well as new methods for optical characterization, paving the way to high-speed super-resolution imaging, real-time detection and manipulation, efficient energy conversion systems and transformative advances in the area of quantum measurements and metrology.

On this path, the photonics community should ultimately build an optical structures and materials genome to construct a dataset of photonic concepts, architectures, components and photonic materials to enable hierarchical machine-learning that could provide ultimate-efficiency devices.

This effort should be extended to realize all-optical platforms to perform deep learning and other AI algorithms at the device level.

of light, ushering in an even brighter AI era.